

USING DIGITAL NUDGES TO ENHANCE COLLECTIVE INTELLIGENCE IN ONLINE COLLABORATION: INSIGHTS FROM UNEXPECTED OUTCOMES¹

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The dramatic expansion of internet communication tools has led to the increased use of temporary online groups to solve problems, provide services, or produce new knowledge. However, many of these groups need help to collaborate effectively. The rapid development of new tools and collaboration forms requires ongoing experimentation to develop and test new ways to support this novel form of teamwork. Building on research demonstrating the use of nudges to shape behavior, we report the results of an experiment to nudge teamwork in 168 temporary online groups randomly assigned to one of four different nudge treatments. Each nudge was designed to spur one of three targeted collaborative processes (collaborator skill use, effective task strategy, and the level of collective effort) demonstrated to enhance collective intelligence in extant research. Our results support the basic notion that digitally nudging collaborative processes can improve collective intelligence. However, to our surprise, a couple of nudges had unintended negative effects and ultimately decreased collective intelligence. We discuss our results using structured speculation to systematically consider the conditions under which we would or would not expect the same patterns to materialize in order to clearly articulate directions for future research.

Keywords: Digital nudges, collaborative processes, collective intelligence, team skill use, team task strategy, collective effort, nudge design, structured speculation

Introduction

There has been a sharp increase in the use of temporary, online groups to carry out work in a variety of settings (Deng, et al., 2016; Michelucci & Dickinson, 2016). Initial forms of online work took place in “crowds,” which were used to carry out

primarily independent or loosely coupled tasks (Afuah & Tucci, 2012; Blohm, et al., 2013) where the management of individuals’ contributions could be electronically mediated effectively (e.g., Kittur, et al., 2011). More recently, there has been a growth in the use of smaller temporary online groups conducting more interdependent work, such as developing

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software or solving problems requiring the integration of specialized knowledge (Retelny et al., 2014; Riedl, et al., 2020; Valentine et al., 2017). While these more interdependent online collaborations also occur on internet-based platforms, the kinds of work involved typically require more fluid coordination than is possible using standard online tools (Chidambaram, et al., 2020; Larson & DeChurch, 2020).

Encouraging a high level of collaboration in temporary, online groups can be difficult given the absence of established roles or norms for working together. Successful team leaders in traditional contexts often use early formative moments to set norms fostering beneficial communication and collaboration patterns for teamwork (Hackman & Wageman, 2005). However, online temporary collaboration contexts typically lack clear roles or opportunities to shape collective norms, and thus the degree to which collaboration develops is dependent on whether one or more collaborators take the initiative to try to influence it.

A possible approach to cue the norms supporting collaborative behavior may exist in behavioral economics research on “nudges.” Nudges are “any aspect of the choice architecture that alters people’s behavior in a predictable way without forbidding any options or significantly changing their economic incentives” (Thaler & Sunstein, 2009). Nudges have been frequently studied in the context of individual buying decisions where a nudge might operate by making certain options more salient (Benartzi et al., 2017; Chapman, et al., 2021) or by establishing an option as the “default” and leveraging common tendencies toward inertia that anchor individuals on that particular choice (Thaler, et al., 2010).

Extending research on nudges into the context of online group collaboration, our central research question here is *how can digital nudges influence the formation of collaborative processes in temporary, online groups?* In our experiment, we tested four digital nudges designed to cue behavior encouraging one of the three collaborative processes shown to increase collective intelligence in extant research (Riedl et al., 2021). We tested the nudges by integrating them into the digital interface of the Test of Collective Intelligence (TCI; Riedl et al., 2021), a test battery that captures the ability of a group to collaborate on a wide range of tasks, a definitional quality of collective intelligence (Woolley, et al., 2010). We randomly assigned temporary online groups to encounter one of our nudges as they worked together on the TCI. Our results provide some support for the general concept of using digital nudges to encourage certain collaborative behavior and increase collective intelligence in this context. However, not all of our nudges had a positive or predicted effect; one of the four had no effect, and two others had *opposite* and negative effects, suggesting additional insights and future directions for research.

Our study contributes to the nascent but growing literature on temporary teams (Retelny et al., 2014; Saunders & Ahuja, 2006) and builds on and extends research on group support systems (GSS) (Ackermann & Eden, 2020) and computer-mediated collaboration (Chidambaram et al., 2020) by examining if and how relatively lightweight “nudges” can be used to shape collaborative processes across a variety of tasks. In discussing our findings, we engage in structured speculation (Banerjee, et al., 2017) to reflect on both our expected and unexpected results to identify some principles to guide future research on the design and use of nudges for collaborative behavior.

Theoretical Background

Collective Intelligence and Collaborative Processes

The use of online groups of problem solvers has exploded in the last few decades. Over time, these have evolved from crowds of relatively independent contributors to smaller, more interdependent temporary “flash” teams (Retelny et al., 2014; Valentine et al., 2017). While, in principle, it can be easier to coordinate teams that are smaller in size (Mueller, 2012), the lack of clear roles or shared norms in temporary online groups can make the collaboration necessary for interdependent work quite challenging (Riedl & Woolley, 2017).

One area of research with potential insights for improving temporary online team collaboration is the literature on small-group performance (Hackman, 1987; McGrath, 1984). This research identifies three essential collaborative processes underlying team performance in a variety of contexts, including the effective use of collaborators’ knowledge and skill, the development of appropriate task coordination strategy, and the generation of high collective effort (Hackman, 1987). These processes have been identified as the major sources of “process loss” in a wide range of collaborative settings (Steiner, 1972). For instance, when groups fail to engage the full effort, motivation, and commitment of all contributors, social loafing or free riding can significantly undercut outcomes (Latane, et al., 1979; Price, et al., 2006). Likewise, when the knowledge and skill of collaborators are not discovered and used to the fullest extent possible, the group cannot achieve its full performance potential (e.g., Argote & Ren, 2012). In addition, the failure to coordinate inputs into a well-designed task strategy can result in costly omissions and errors (Aggarwal & Woolley, 2013; Hackman, et al., 1976).

Building on this work, recent research demonstrates that these three collaborative processes—skill use, task strategy, and collective effort—are significant drivers of collective

intelligence in both face-to-face and online teams, whether members are strangers or have an ongoing working relationship (Riedl et al., 2021). Furthermore, research on interventions to improve these collaborative processes has demonstrated that they are important leverage points for altering the trajectory of a group's work (Hackman & Wageman, 2005). Studies of such interventions have shown that even short, targeted efforts can improve collaborative processes and have a substantial impact on performance, particularly when they occur early in a group's life (Fisher, 2017; Hackman & Wageman, 2005; Mathieu & Rapp, 2009).

An important unanswered question is whether an intervention administered by technology rather than an experienced human coach can be effective enough to improve collaborative processes and enhance collective intelligence. Research on group support systems (GSS), an active area of research in the 1990s, demonstrated some benefits associated with the use of electronic tools for structuring specific activities in teams, with some studies observing better engagement, higher creativity, and improved information-sharing across members resulting from tool use (Chidambaram & Jones, 1993; Nunamaker, et al., 1996) although not uniformly (e.g., Dennis, 1996). Such systems were typically developed and optimized for particular task types (such as brainstorming or decision-making) rather than to support more general collaborative processes that would be relevant to a range of different tasks a group may undertake (Chidambaram et al., 2020). More recent tools used to facilitate online collaboration across a range of tasks or projects tend to be more general and typically focused on communication (e.g., Slack) or designed to facilitate distributed taskwork without particular features to structure specific collaborative processes (e.g., G Suite). Here, we build on and complement extant research and practice by examining whether lightweight digital interventions into specific collaborative processes can effectively improve performance across a range of different task types.

The Potential for Digital Nudging to Improve Collaborative Processes

In the abstract, it might seem possible to design a system that would completely manage online collaboration to ensure that contributors interact in ways known to result in better team outcomes. However, extant research has repeatedly demonstrated that the use of and compliance with even the smartest technology system is far from assured (e.g., Dietvorst, et al., 2015; Glikson & Woolley, 2020; Kane & Labianca, 2011). Users are particularly likely to distrust or ignore algorithmic advice related to social or interpersonal situations (Hancock et al., 2011), particularly when they have no control over or insight into the basis of its guidance or decisions (Glikson & Woolley, 2020; Rahman, 2021).

Since most humans resist being controlled by other humans or algorithms, recent research suggests that a more effective approach to influencing behavior while preserving autonomy may be possible through using a “nudge.” A noteworthy feature of nudges is that they operate without forcing or constraining an individual's decision; they direct attention to certain options in a choice environment, raising the probability of a particular outcome while leaving all options available to be freely chosen (Thaler, 2021). Nudge principles have also been applied to digital choice environments; digital nudging occurs via “the use of user-interface design elements to guide people's behavior in digital choice environments” (Weinmann, et al., 2016, p. 433). Compared to offline contexts, digital nudging offers the additional intriguing possibility of customizing the form and timing of a nudge based on an individual's prior decisions or behavior, thereby potentially having an even greater impact than traditional nudges (Mirsch, et al., 2017).

Extant research on nudges has identified a set of basic principles by which they operate, founded on well-understood cognitive mechanisms (Mirsch et al., 2017). One of the most commonly used nudge principles is “set defaults” (Thaler et al., 2010). In its basic form, the default is the decision that results if the user does nothing to change it, such as the “standard” installation when installing a new software package. Defaults can serve to invoke status quo bias (Mirsch et al., 2017), in that they require more cognitive effort to get users to search for alternatives to make a change to the situation as presented. The default can also operate via social norms in that users may assume that the default exists because it is what most people choose. An example of how a “setting defaults” nudge might be used to shape collective behavior would be structuring a workspace where tasks are presented in a particular numbered order or pre-assigned to different individual group members, which would lead most groups to start with “Task 1” or to proceed by doing the tasks designated for them.

Another nudge principle that could be useful in a collaborative environment is “give immediate feedback” (Thaler et al., 2010). This nudge works to shape individual behavior by providing relevant comparison-oriented information soon after a decision is made or a particular behavior is engaged in, serving to change perceptions and social norms (Mirsch et al., 2017). This could be useful in temporary online collaboration settings, where perceptions that others are social loafing or free riding can quickly lead contributors to reduce their own contributions in response and undermine the collaboration overall. A nudge that gives feedback and makes each collaborator's level of contribution visible could help alleviate such concerns by evoking social norms to equalize contributions, as recent research on distributed student groups has demonstrated (Glikson, et al., 2019).

A third nudge principle that could be helpful in a collaborative environment is “structure complex choices” (Thaler et al., 2010). When faced with a large and potentially overwhelming set of unstructured decisions, individuals tend to use simplifying heuristics, which risk emphasizing salient but less important attributes as a means of narrowing options. Nudges designed to structure complex choices commonly do so by organizing options or sequencing decisions so that choices are guided by the most important criteria, increasing the probability of a better outcome. Collaboration often involves a set of complex and interrelated individual and collective decisions, where early decisions on seemingly minor points can be more consequential than collaborators realize. A nudge based on this principle could take the form of a facilitator that prompts discussion about essential issues that might not be discussed otherwise (e.g., What is each person’s skill set? Who is available when? How should we allocate tasks?) Cueing such discussion can result in a major improvement over the more common scenario in which each person just picks something to work on with little to no discussion of any related topics.

These examples illustrate the potential for nudges to improve collaborative behavior. Extant research has shown that in the absence of any prompting, most groups neglect explicitly discussing coordination; however, when prompted to spend even a few minutes doing so, especially at the beginning of their work, even short conversations can yield great benefits for performance (Hackman & Wageman, 2005; Shuffler, et al., 2018). Therefore, the right kind of digital nudge at the right moment could have a substantial impact on CI.

That said, this new frontier of technology-driven management is relatively uncharted and fairly complex (Kellogg, et al., 2019). While nudges have been successful in influencing private individual decisions, a group’s response to a stimulus is a complex result of how individuals respond, each person’s perception of how others react, and how those responses and perceptions affect each other. Consequently, it can be difficult to predict the effects of even a simple nudge in the complex adaptive system comprising collaborative groups (Arrow, et al., 2000).

To explore whether digital nudges can enhance collaboration and collective intelligence in temporary, online groups, we designed a set of nudges targeting specific collaborative processes and tested them in an online experiment. Figure 1 illustrates the conceptual model guiding our research.

Methods

Participants

We recruited 515 Amazon Mechanical Turk workers who were randomly assigned to one of 168 three- or four-member groups. Each group was randomly assigned to one of the four digital nudge treatment conditions (described below) or the control condition ($n = 37$). The participants were paid a fixed rate individually for participating in a session lasting up to 45 minutes, with no performance-based bonus. The sample was 39.6% female and participants’ average age was 35.6 years old ($SD = 10.1$).

Tasks and Procedure

Experimental Procedure and Platform.

The experiment was administered via the Platform for Online Group Studies (POGS; pogs.wiki), a browser-based platform that facilitates synchronous collaboration. As shown in Figure 2, the POGS interface displays task instructions on the left and a collaborative workspace in the center of the screen where all members can view and input text synchronously. On the right panel, collaborators can communicate with each other via chat. During the experiment, collaborators worked via POGS on the Test of Collective Intelligence (TCI), an online task battery designed to measure collective intelligence (Riedl et al., 2021). After finishing the TCI, participants completed a survey individually that included measures of demographics and social perceptiveness along with experimental manipulation checks.

Digital Nudge Treatments

Each of the digital nudges was tested in one of four experimental conditions, deployed as groups completed the TCI, with a fifth “no nudge” control condition as a comparison. Table 1 summarizes the focus of the four nudges, which are illustrated in Figure 2.

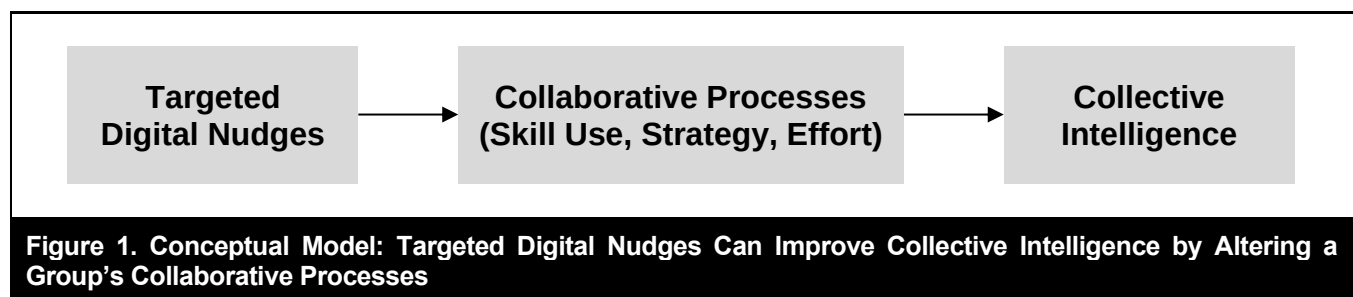
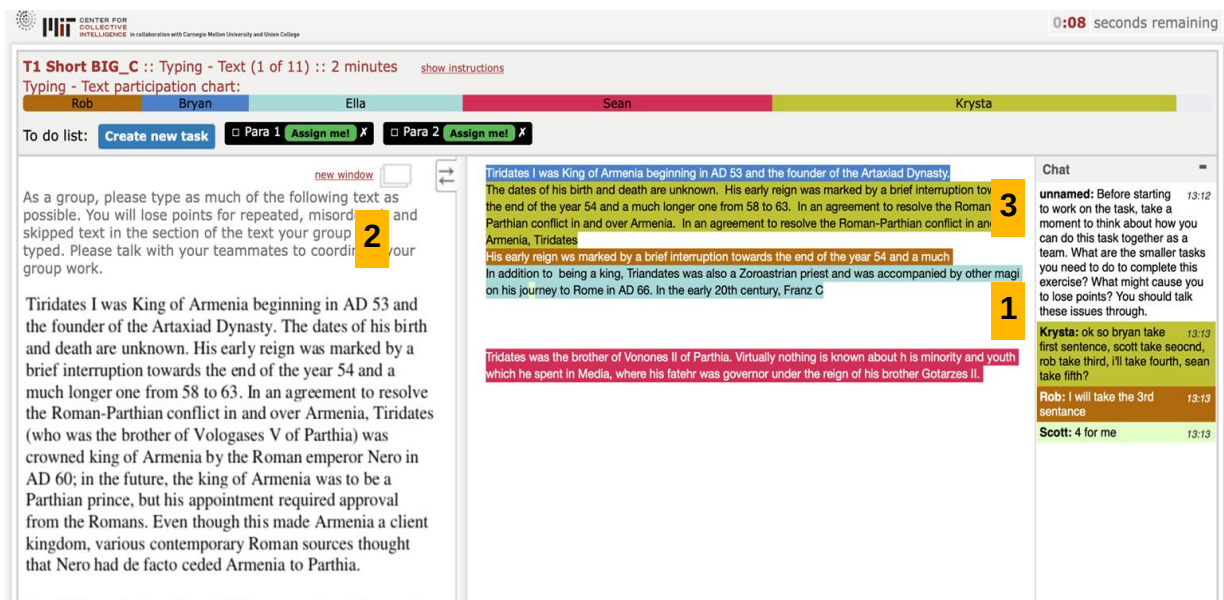


Table 1. Description of Digital Nudges Used in the Study

Digital nudge	Target collaborative process	Nudge design principle applied	Description
Skill Facilitator Bot (n = 29)	Skill use	Structure complex choices	This chatbot initiated consideration of member knowledge and skill. At the beginning, the facilitator would ask things such as "Who is good at [typing/word problems/skill relevant to task]?"
Strategy Facilitator Bot (n = 33)	Task strategy	Structure complex choices	This chatbot prompted collaborators to discuss if or how to divide the work and agree on what each contributor would do at the beginning of new tasks. At the midpoint, it prompted the group to review their task coordination strategy to see if they needed to make adjustments.
ToDo List (n = 33)	Task strategy	Set defaults	This tool was present at the beginning of work and available throughout, intended to prime collaborators to think about and discuss task assignments, identify gaps, and resolve any conflicts, and check off things as completed to update progress.
Feedback Display (n = 36)	Effort	Provide immediate feedback	This real-time indicator of each collaborator's relative effort was present throughout their work, based on their activity level on the task platform

Note: Participants in each nudge treatment condition were exposed to only one nudge type. All nudges were presented as described during each task in the TCI.



Note: Participants in each treatment condition saw only one of the following: (1) Questions to initiate discussion by the *Skill Facilitator Bot* or the *Strategy Facilitator Bot* were shown in the chat window on the right to participants in the skill use and task strategy conditions, respectively. (2) An interactive task management tool was available at the top of the screen to participants in the *To Do List* condition. (3) A dynamically updated display of relative participation was visible at the top of the screen to participants in the *Feedback Display* condition. All digital nudges were disabled for control condition participants.

Figure 2. User Interface of the Test for Collective Intelligence (TCI) Depicting the Four Targeted Digital Nudges

Following extant research on team interventions (Hackman & Wageman, 2005; Mathieu & Rapp, 2009), the nudges were first deployed just as teams began working together. As described in Table 1, each nudge was based on the nudge principle that seemed most appropriate for the targeted collaborative process. For instance, both *skill use* and developing an appropriate *task strategy* in a team involve consideration of a complex set of inputs. Thus, we designed facilitator bots, drawing on the principle of “structure complex choices.” Each facilitator bot prompted a discussion of relevant details; specifically, the *Skill Facilitator Bot* asked about member task skills and the *Strategy Facilitator Bot* focused on the coordination of taskwork. We also tested an alternative nudge design task strategy based on the principle “set defaults.” To implement this, we placed a *ToDo List* widget in the workspace to prime collaborators to think about task assignments early in their work, reasoning that by making coordination salient, collaborators would be more likely to discuss task strategy.

Finally, the fourth nudge was based on the principle “provide immediate feedback” and displayed real-time information on each collaborator’s level of effort at the top of the user interface through their work. We reasoned that making this visible would reduce concerns about free-riding and encourage those contributing less than others to increase their effort.

In designing the nudges, each nudge went through multiple rounds of engineering tests and debugging. This was followed by approximately 4-5 pilot test groups with naive participants to ensure that each nudge functioned as expected and participants understood how to use it. After completing this process with each of the four nudges, we conducted our full experiment.

Measures

Collaborative Processes.

As groups collaborated on the TCI, the three collaborative processes that we targeted with the digital nudge treatments were measured using participant activity data captured in POGS. The level of collective effort (*effort*) was based on the total activity level of collaborators, calculated by aggregating the number of keystrokes made by all contributors during the group’s work. The coordination of *task strategy* was operationalized by calculating the total percentage of each group of tasks that was attempted or completed, as a common indicator of poor coordination on the TCI is forgetting or leaving parts of tasks incomplete. Finally, the use of collaborators’ skills (*skill use*) was operationalized by first estimating the skill of each contributor on each task based on the quality of independent inputs and then calculating the task-

wise correlation of this skill index with each collaborator’s relative effort on each task, resulting in a score ranging between -1 and 1.

Collective Intelligence

The *collective intelligence (CI)* for each group was based on their score on the TCI. The TCI is a 25-minute task battery sampling a variety of collaborative task types based on existing group task taxonomies (i.e., decision-making, idea generation, task execution, etc. [Larson, 2010; McGrath, 1984] see tci.wiki for more details). The TCI provides a robust metric of a group’s ability to work together across a range of tasks, where over 100 different inputs are combined to calculate the CI score. A recent meta-analysis of 22 datasets, including over 1,300 groups, supports the construct validity of the TCI (Riedl et al., 2021). The reliability of the TCI measure in this sample is Cronbach’s $\alpha = 0.918$. The TCI was administered via POGS, which managed the delivery of instructions and the amount of time allowed for work on each task (ranging from 2-6 minutes) while synchronizing and displaying collaborators’ inputs in real time.

Control Variables

In all analyses, we controlled for group size, average member social perceptiveness (as measured by the Reading the Mind in the Eye test, see Baron-Cohen, 2000), as well as amount and variance in communication (measured via text chat), as these have been shown to predict CI in prior studies (Engel, et al., 2014; Woolley et al., 2010).

Results

Manipulation Check

Before analyzing the effects of our nudge treatments, we first conducted manipulation checks to ensure that participants noticed or responded to each nudge in a manner that met at least a minimum threshold for interpretation. All participants completed a post-task survey in which they were asked if they noticed the applicable nudge.

For the *ToDo List* and *Feedback Display* conditions, 85.15% and 87.39% of participants in each condition, respectively, reported noticing the relevant feature on their dashboard, with a minimum of at least one member of every single team. In addition, for the *ToDo List*, we observed that all but five of the teams interacted with the tool at least once; of the teams who did not use it, most or all of the members reported noticing the tool on the survey.

The *Skill* and *Strategy Facilitator Bot* nudge treatments both prompted collaborators to communicate with each other, such that if a team responded in any manner, their level of communication should exceed that of the control condition, which received no nudge treatment. Consistent with this expectation, we observed that in both the *Skill Facilitator Bot* ($t_{(64)} = 3.46$, $p = 0.0007$, one-tailed) and the *Strategy Facilitator Bot* ($t_{(68)} = 1.77$, $p = 0.0387$, one-tailed) conditions, teams communicated significantly more than teams in the control condition.²

Analysis Approach

We use a bootstrapping-based PROCESS model for multiple mediation analysis (Process Model 4; Hayes & Preacher, 2014) to test the effects of our four digital nudge treatments (with control condition as a baseline) on the three collaborative processes (skill use, task strategy, and level of effort), and their effect on collective intelligence. This analysis used a multi-categorical independent variable to capture the independent effects of each treatment (using OLS regression) and estimated the relative mediation effect of the three processes acting in parallel (via bootstrap sampling), providing a more powerful analysis compared to running a separate regression for each nudge-process pair independently (Hayes & Preacher, 2014). The analysis was carried out using the *pyprocessmacro* library (Andre, 2017/2020) in Python 3.0. The descriptive statistics and correlations for study variables are shown in Table 2.

Test of Experimental Nudges' Independent and Indirect Effects

The results of our analysis of the effects of the four nudges on their targeted collaborative processes, as well as their independent and indirect effects on collective intelligence, are shown in Tables 3a (independent effects) and 3b (indirect effects) and represented graphically in Figure 3. Consistent with extant research, we first observed that the degree to which teams exhibited all three collaborative processes is associated with higher collective intelligence ($R^2 = 0.63$, with positive coefficients for all three processes at $p < 0.05$; Table 3a, Model 4). However, to our surprise, not all the nudges had the intended positive effect on their targeted process (Figure 3).

(1) As expected, *Skill Facilitator Bot* significantly improved skill use as intended (Table 3a, Model 3) and, in doing so, exhibited a significant indirect effect on collective

intelligence ($b = 0.044$, 95% confidence intervals = {0.002, 0.132}; Table 3b, Model 3).

(2) However, *Strategy Facilitator Bot* did not have a significant effect on task strategy as intended (Table 3a, Model 2) nor an indirect effect on CI ($b = -0.128$, 95% confidence intervals = {-0.371, 0.080}; Table 3b, Model 2).

(3) The *ToDo List* nudge, unexpectedly, resulted in a statistically significant weaker task strategy (see Table 3a, Model 2) and a negative indirect effect on CI via task strategy ($b = -0.267$, 95% confidence intervals = {-0.540, -0.052}; Table 3b, Model 2).

(4) The *Feedback Display* nudge, also unexpectedly, had a significant negative effect on the total level of effort (see Table 3a, Model 1) and a negative indirect effect on CI via effort ($b = -0.132$, 95% confidence intervals = {-0.326, -0.019}; Table 3b, Model 1).

Exploring the Unexpected Results

Given the null effect of the *Strategy Facilitator Bot* and the surprising negative effects of the *ToDo List* and *Feedback Display* nudges, we examined more details about participants' collaboration behavior as well as open-ended post-task survey responses to gain additional insight into why this occurred and to guide future research.

Strategy Facilitator Bot

Participants commented in post-task surveys that at first, they found this facilitator helpful. However, over time, as it became clear that the extent to which the facilitator could help them was more limited than they had initially expected, they tuned it out. Consequently, during the final TCI task that included two minutes of planning time for all teams, the *Strategy Facilitator* teams communicated significantly less compared to all other conditions (contrast comparing the amount of chat discussion to all other conditions: $F_{(1,163)} = 3.96$, $p = .048$). This pattern, where initially high expectations gave way to a steep decline in engagement with the bot when those expectations were unmet, likely explains the lack of any statistically significant effect. This is consistent with patterns noted in recent studies of trust development between humans and virtual agents (Glikson & Woolley, 2020) and points to an important caveat for future work on deploying what are intended to be "simple" nudges.

² As an additional robustness check on our results, we conducted our analyses both with and without the teams that had one or more members that did not report awareness of the nudge, or who engaged in less chat than

the 50th percentile of the control condition. All results support the same conclusions whether analyses include or exclude these teams.

Table 2. Item Descriptives and Correlations for Study Variables

Variable	M	SD	min	max	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Collective intelligence	-0.04	0.94	-1.87	2.60	.68*	.74*	.15	.17*	.37*	-.02	.07
(2) Effort	0.49	0.09	0.30	0.78	-	.76*	.04	.26*	.19*	.09	.09
(3) Task strategy	0.43	0.08	0.26	0.66		-	.08	.23*	.15	-.10	.03
(4) Skill use	0.77	0.19	-0.08	1.00			-	.09	-.05	-.11	.08
(5) Group size	3.07	0.25	3	4				-	.01	.03	.06
(6) Social perceptiveness	25.65	4.18	15.67	33					-	.27*	.13
(7) Communication amount	48.89	0.49	3	152						-	.62*
(8) Communication variance	5.01	3.89	0	18.01							-

Note: $N = 168$ groups. $*p < .05$

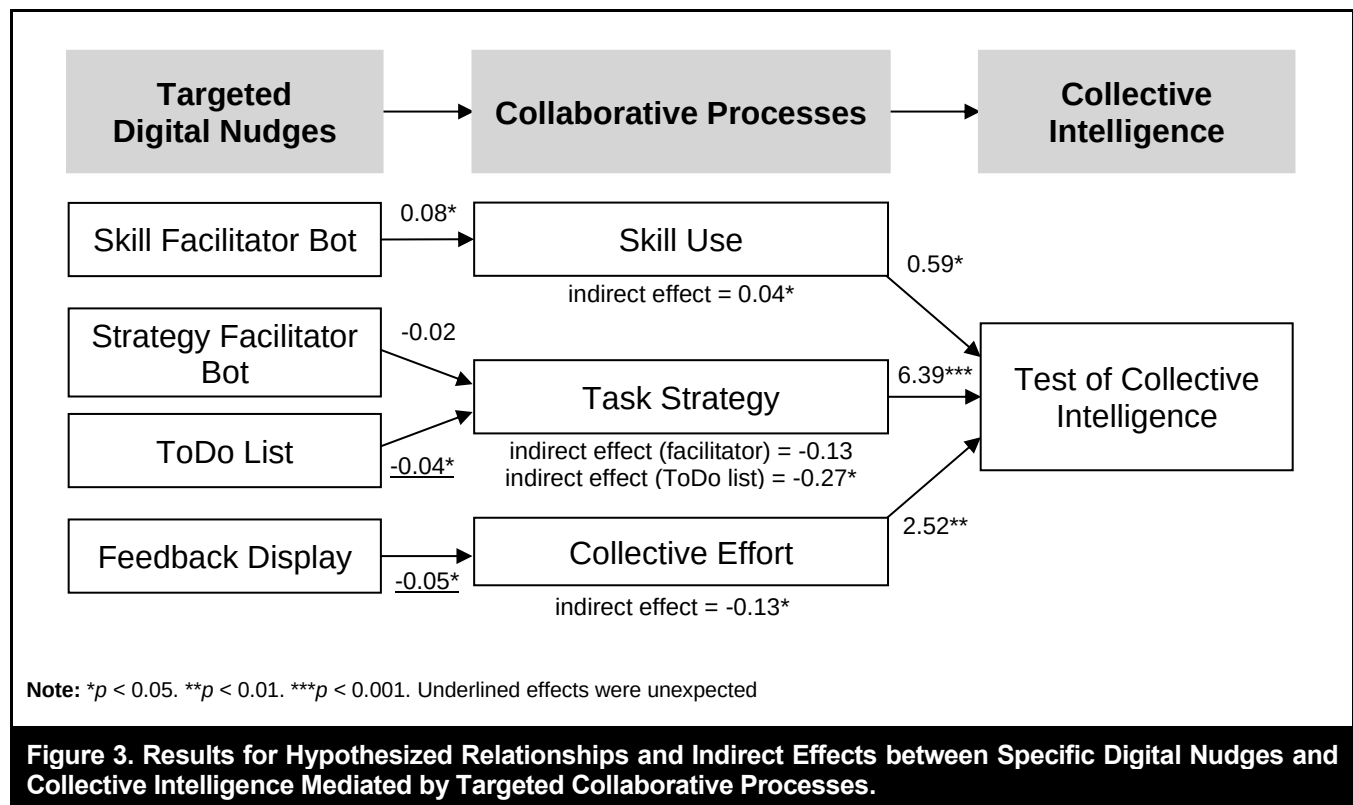


Table 3a. Results of Hypotheses Testing: Outcome Regression Models estimated using OLS in PROCESS Model 4 showing the Effects of Different Nudges on Collaborative Processes and Collective Intelligence

	Level of collective effort (Model 1)		Appropriate task strategy (Model 2)		Appropriate skill use (Model 3)		Collective intelligence (Model 4)	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Constant	0.134	0.088	0.181 *	0.082	0.657 ***	0.166	-5.805 ***	0.604
<i>Skill Facilitator Bot</i>	-0.019	0.022	-0.026	0.019	0.076 *	0.045	0.185	0.166
<i>Strategy Facilitator Bot</i>	-0.029	0.021	-0.02	0.018	-0.016	0.046	0.14	0.138
<i>ToDo List</i>	-0.011	0.022	-0.042 *	0.018	-0.084	0.054	0.332 *	0.131
<i>Feedback Display</i>	-0.052 *	0.023	-0.071 ***	0.019	-0.024	0.045	0.289	0.155
Team size	0.095 ***	0.023	0.077 **	0.023	0.073	0.039	-0.141	0.184
Social perceptiveness	0.003 *	0.002	0.003	0.001	-0.001	0.004	0.064 ***	0.012
Comm. amount	-0.001	0	-0.001 **	0	-0.002 **	0.001	-0.002	0.002
Comm. variance	0.002	0.002	0.004 *	0.002	0.013 *	0.005	0.002	0.002
Skill use							0.587 *	0.235
Task strategy							6.393 ***	1.006
Effort							2.521 **	0.852
<i>N</i>	168		168		168		168	
Adj. <i>R</i> ²	0.103		0.157		0.066		0.637	

Note: **p* < 0.05. ***p* < 0.01. ****p* < 0.001. Gray-shaded cells are the coefficients used to show mediation in Figure 3.

Table 3b. Results of Hypothesis Testing: Direct and Indirect Effects of Different Nudges on Collaboration Processes and Collective Intelligence estimated using Bootstrap Sampling in PROCESS Model 4

Condition	Indirect effect via effort (Model 1)	Indirect effect via task strategy (Model 2)	Indirect effect via skill use (Model 3)	Direct effect on collective intelligence (Model 4)
<i>Skill Facilitator Bot</i>	-0.047 CI (-0.197, 0.050)	-0.166 CI (-0.438, 0.045)	0.044* CI (0.002, 0.132)	0.185 CI (-0.139, 0.510)
<i>Strategy Facilitator Bot</i>	-0.073 CI (-0.217, 0.016)	-0.128 CI (-0.371, 0.080)	-0.009 CI (-0.081, 0.039)	0.14 CI (-0.131, 0.410)
<i>ToDo List</i>	-0.028 CI (-0.150, 0.077)	-0.267* CI (-0.540, -0.052)	-0.05 CI (-0.159, 0.002)	0.332* CI (0.074, 0.589)
<i>Feedback Display</i>	-0.132* CI (-0.326, -0.019)	-0.454* CI (-0.758, -0.210)	-0.014 CI (-0.088, 0.033)	0.289 CI (-0.015, 0.592)

Note: *N* = 168 groups. **p* < 0.05. CI: 95% confidence interval. Gray-shaded cells highlight the indirect effect of the collaborative process specifically targeted by the nudge in that condition. These statistics are also included in Figure 3.

ToDo List

Interestingly, while our ToDo list resulted in a *weaker* task strategy, at the same time, it apparently *enhanced* collective intelligence (Table 3b, Model 4; direct effect). In examining the score distributions more closely, we note that none of the teams receiving the *ToDo List* nudge performed as badly as the lowest-performing teams in the other conditions. We attribute this to the fact that teams exposed to the *ToDo List* nudge engaged in more communication early in their work. During the first task, *ToDo List* teams communicated significantly more than control groups ($t_{(68)} = 2.08, p = 0.04$), and in examining the content of their communication, we see that these early chats were almost entirely focused on task strategy. We suspect that this may have had the effect of improving the work of the teams that would have been very low-performing otherwise—similar to the teams documented in many prior studies who engage in little to no discussion or coordination unless specifically prompted to do so, reliably producing very weak outcomes (Hackman et al., 1976; Hackman & Wageman, 2005).

At the same time, there were definitely other teams that found the widget distracting or too arduous to be helpful. As one participant commented: “I used it once to test it out, but it didn’t really seem that useful to me. Time spent creating a ToDo could have been spent trying to do the actual task.” Consequently, we observed a bifurcated effect, where the *ToDo List* nudge prompted task strategy coordination in teams that were unlikely to engage in any coordination otherwise, thus scaffolding the processes of lower-performing teams and raising the “floor” in this condition. This explains the higher collective intelligence on average. Unfortunately, without a means of mitigating a preoccupation with task coordination, most teams that used the widget spent too much time planning and not enough time doing the work they planned, running out of time on many tasks and exhibiting poor task strategy overall, compared to other conditions.

Feedback Display.

To unpack how collaborations responded to the feedback display, we examined the detailed behavioral collaboration data from teams and their comments on post-task surveys. First, we noted that the *Feedback Display* nudge elicited strong and somewhat polarized reactions. Some participants were positive, telling us: “I liked that the chart showed who was contributing,” and “It let me know if we were working fairly,” as well as “Great idea to get everyone to work.” In contrast, others expressed: “I did not like it much. It created a feeling of anxiety and inadequacy if a team member underperformed,” and “A

cooperative event should never be competitive,” as well as “IT JUST PUT PRESSURE ON ME [sic].”

In trying to better understand how these different responses manifest in the level of effort contributed by different members, we compared the amount of effort of the highest- and lowest-effort contributors in the *Feedback Display* and control conditions. We noted that the highest-effort contributors in both conditions were working equally hard ($t_{(71)} = 1.42, p = 0.159$). However, the lowest-effort contributors in the *Feedback Display* condition contributed less than the lowest-effort contributors in the control condition: ($t_{(71)} = 3.14, p = 0.005$).

We further examined the gap in effort between the highest- and lowest-effort contributors in each group during different time phases of the groups’ work. We observed that a gap emerged between the highest- and lowest-effort contributors in the *Feedback Display* groups following the very first task of the TCI that was significantly larger than the gap observed in the control condition: ($t_{(71)} = 1.99, p = 0.025$, one-tailed, which was sustained throughout their work together. We interpret this pattern as suggesting that the polarized response to this nudge evident in comments on the post-task survey sparked some reactance in participants that crystallized into a pattern of more variation in effort in those teams. Consequently, rather than invoking social norms encouraging more equal contributions to collective work as intended, the focus of this nudge on comparing the relative effort of collaborators to each other was perceived as “public shaming” and led some to respond by withdrawing effort completely.

Discussion: Structured Speculation

We have reported the results of an experiment to test the use of digital nudges to subtly cue collaborative behaviors in temporary online teams to increase collective intelligence. Consistent with extant research, we observed that the collaborative processes predicted collective intelligence. However, as reported, not all of our attempts were “successful” in the sense that they resulted in behaviors and outcomes that were unexpected—in two cases, the *opposite* of what was anticipated.

We offered some potential explanations in our exploratory analysis, and we discuss them further here, using “structured speculation” (Banerjee et al., 2017) to systematically consider the conditions under which we expect that researchers would or would not observe the same pattern of results.³ Structured

³ We are deeply indebted to the associate editor who handled this paper, whose incredibly insightful feedback resulted in many improvements throughout the review process, including the suggestion to include this

“structured speculation” in order to systematically surface specific insights as possible to guide future research.

speculation can help researchers explicitly articulate the boundary conditions of their results by considering how issues of *context*, *scale*, and *population* might change the results. Doing so is important for enabling other researchers and practitioners to understand the extent to which the findings of a study will generalize to other studies or applications.

In our structured speculation, we build upon our exploration of these unexpected findings to discuss what can be learned about nudge design for teams and how generalizable these findings are. Specifically, we consider (1) *Context*: Would the same nudge treatments have a different effect depending upon our choice of nudging target? For instance, our treatment design choices were informed by nudge principles that target individual cognitive processes. In their current form, do these principles transfer *equally* to the design of nudges for collective-level collaborative processes studied in this context? (2) *Scale*: From a dose-response perspective, how does changing the intensity of nudging affect the target processes differently? (3) *Population*: Would the same treatment have the same effect on a different population? Or under what circumstances will the nudges work for digital teams? See Table 4 for an overview.

Context: Individual- vs Collective-Level Processes as Nudge Targets

We designed our nudges based on principles derived from research on individual decision-making, selecting those that were connected conceptually with each of the collaborative processes that extant research on collective intelligence shows are essential (e.g., Riedl et al., 2021). One question is whether each nudge was designed to be successful in the context of *collective* behavior. The results for several of our treatments suggest that this is not the case.

For instance, the *ToDo list* was based on the principle of “setting defaults” but did so in a manner that cued individual members to focus on their own individual tasks rather than the collective coordination of tasks across the team. Similarly, the *Feedback Display* condition was based on the nudge principle of “providing immediate feedback” but focused on comparing individuals’ contributions to each other, provoking interpersonal comparisons between members that led to some strong negative reactions. An alternative design to make this a more effective nudge in the context of collective behavior might be a display highlighting collective effort, aggregated and undifferentiated. Perhaps compared to a relevant external benchmark (such as the average effort of other teams) or

create an explicit anchor to a desired *collective* goal state (e.g., only 200 words to go). The potential for such an approach to be more effective is supported by other studies of online collaboration demonstrating that group-based goals can be more effective at increasing members’ motivation than individual-level goals (Ling et al., 2005).

By contrast, our successful nudge was the *Skill Facilitator Bot*, based on the principle of “structure complex choices.” This nudge prompted team members to share their task expertise with one another, thereby helping team members get to know each other and the resources available to the group. Since each task involved different skill sets, each iteration of this discussion yielded new information and contributed to the formation of the team’s shared cognition, a resource extant research demonstrates is essential to teamwork (DeChurch & Mesmer-Magnus, 2010). Importantly, there is evidence suggesting that these benefits of transactive memory are disrupted if the process of learning about each other is automated in service of lower individual cognitive load (Gupta & Woolley, 2018). Thus, in addition to prompting coordination-related thoughts in individual members, the nudge also initiated collective discussions among members that facilitated improved coordination over time.

Taken together, we postulate that nudges that draw attention to collective goals and resources and explicitly cue coordinative actions are more likely to succeed, compared to nudges that cue closed-loop individual cognition and behavior that evokes comparison or competition within teams. This inference extends our conceptual understanding of nudging collaborative teamwork and provides information system designers with a “rule of thumb” for making and evaluating their digital design decisions.

Scalability: Effects of Nudging Team Processes from a Dose-Response Perspective

When considering the scalability of an intervention, the concern is whether the effects observed would change if the intensity of nudging (dosage) of the targeted processes were different. A typical dose-response relationship characterizes a medicinal drug’s efficacy. Setting too high a dose is unsafe for the patient while selecting too low a dose makes it difficult to validate the drug’s usefulness. Analogously, we question if it is possible that the dosage of the nudges we design, in terms of their characteristic features and temporal presence, to guide team interactions have a similar dose-response relationship.

Table 4. Structured Speculation on the Generalizability of Study Results to Other Settings

	Context: Level of processes targeted	Scalability: Dose-response effect	Population attributes		
			Team type	Member type	Member incentives
Skill Facilitator Bot	✓ collective ✗ individual	✓ dose tested in chat—works well	✓ temporary ✗ pre-existing	✓ anonymous ✗ familiar	✓ team performance ✗ individual performance ✗ individual effort
Strategy Facilitator Bot	✓ collective ✗ individual	✗ dose tested in chat—too low	✓ temporary ✗ pre-existing	✓ anonymous ✗ familiar	✓ team performance ✗ individual performance ✗ individual effort
ToDo List	✗ collective ✓ individual	✗ dose tested as dynamic widget—too high	✓ temporary ✓ pre-existing	✓ anonymous ✓ familiar	✓ team performance ✗ individual performance ✗ individual effort ✓ reputational stake
Feedback Display	✗ collective ✓ individual	✓ dose tested as dynamic visuals—works well	✓ temporary ✓ pre-existing	✓ anonymous ✓ familiar	✓ team performance ✗ individual performance ✗ individual effort ✓ reputational stake

Often, the initial focus in studying treatment effects is whether or not a strong enough “dose” of treatment was used for a response to develop. However, doses that are too large can also be problematic, as they can lead to unintended and harmful side effects. Another issue in determining the right dose of a treatment is the development of “tolerance,” where a dose of a treatment that was effective initially becomes ineffective (Bespalov, et al., 2016). In thinking about these issues in the context of our nudge treatments, we question whether some nudge designs were ineffective because the dose tested was too low. Alternatively, if some nudges were delivered in too high a dose or too frequently, did some of our participants develop a form of “tolerance” that led them to ignore the treatment, making it ineffective? Given the dynamic and emergent nature of team processes, we speculate that both of these explanations could apply to some of our nudges.

The results for both of the nudges that we designed to improve a team’s task strategy can be plausibly reinterpreted from this perspective. In exploring our observations of collaborators’ behavior, we noted that a number of groups appeared to eventually ignore our *Strategy Facilitator Bot*—which we interpreted as a suggestion that the “dose” was too low. Since this pre-scripted bot asked the same question at the beginning of each task (i.e., “Who wants to do what?”), while neither providing nor eliciting any new information, it is likely that participants quickly found out that it wasn’t smart enough to be helpful and tuned it out. A slightly different dose-dependent dynamic seemed to occur with *ToDo List*, which was ever-present throughout the collaborators’ work. While some participants quickly began to ignore it, others apparently engaged with it too much (the dosage was too high), such that

it displaced other important processes, such as carrying out the actual work on the task that was planned.

By comparison, one potential reason the *Skill Facilitator Bot* appears to be more successful is that the responses it elicited from participants varied more; each discussion the bot prompted revealed new and useful information that enhanced shared cognition and the quality of collaboration. Thus, the dose tested for that nudge seems to be at the correct level, as it remained effective throughout the testing session.

A step toward resolving this issue of “tolerance” would be to create adaptive or artificially intelligent systems capable of presenting nudges and regulating their dosage (intensity or time) based on how the team processes respond in real time. For example, in a future study, an intelligent nudge system could monitor the rate at which the team is working (e.g., task burndown rate) and nudge its members to reconvene and discuss strategy when it encounters a drop or stagnation in the rate of work completion. Research on such responsive and intelligent tools is already spanning multiple domains relevant to collaborative work. Systems that scale and coordinate analogical reasoning across crowds (Kittur et al., 2019) or scaffold scientific argumentation (Clark & Sampson, 2007) could be instrumental in enhancing collective strategizing and decision-making.

Building on these observations, we postulate that adaptive nudging systems will yield significantly better results in collaborative contexts than the static design approaches that are prevalent (and effective) in nudging individual cognition. In

practice, this underscores the need for developers to not only design feature-rich collaboration tools but also to carefully study and test the psychological and social effects of their choices on team processes. This means evaluating designs not just for functionality and usability but also for their overall impact on collaboration. The unexpected findings of our study act as a cautionary tale—even simple and well-intentioned digital tools can have unintended psychological effects and may cause more harm than benefit to collaboration.

Population: Different Circumstances under Which These Nudges Work for Digital Teams

In our experiment, we tested the effects of digital nudging on collaboration processes and collective intelligence in temporary teams composed of anonymous strangers who were paid for their time. Will the nudges work for preexisting groups where members know each other? How would their effect differ if the members were incentivized differently?

“Blank slate” teams that are temporary, with members who are unfamiliar with one another and have no preexisting norms or routines that have to be altered or preexisting relationships that set the baseline for interaction are likely the most open to nudges capable of shaping their norms of collaboration. Such interventions should be particularly effective during the early phase of their time together (Hackman & Wageman, 2005). In contrast, preexisting teams, where members have had a chance to work with each other and have already agreed upon relatively strong norms of interaction are unlikely to be easily nudged.

However, while high-performing teams with existing norms may be unlikely to be nudged, one cannot assume that all teams with existing norms are high performing. Dysfunctional teams with strained relationships and poor collaboration norms are just as likely to emerge—if not more so. Indeed, such teams may stand to benefit the most from digital interventions that nudge their collaboration processes in real time. We speculate that the success of nudges in preexisting teams is likely to be contingent on the level of trust or psychological safety (Edmondson & Daley, 2020) that emerges within the digital team. The ability to communicate without the fear of judgment, or psychological safety, precedes all collaborative processes.

A future study could test this contingency model by subjecting both preexisting teams and stranger teams to these nudges as they complete the TCI and observing the pattern of interactions of the team's performance with their pre-task measure of psychological safety. We would expect that preexisting teams with higher levels of psychological safety would show the highest performance.

Specifically, regarding stranger teams, the anonymous aspects of such an environment can make it especially difficult to build trust or even the minimum level of commitment necessary to invest a significant amount of effort in task coordination. Further, anonymous collaborators are likely to engage in unhelpful behavior, including social loafing or free riding, which is difficult to address, as we observed in our failed attempts here. However, in the case of preexisting teams, high levels of familiarity with each other developed during previous work contexts can obviate the value of both the *Skill Facilitator Bot* and the *Strategy Facilitator Bot* while still deriving value from the more dynamic nudges, *Feedback Display* and *ToDo List*.

Another contributing factor is the participants' level of motivation, as driven by their incentives for participation. In this experiment, compensation was not performance based, so we can only speculate how participants' responses might have been different if they could have earned more money based on their team's performance. While team performance incentives tend to increase individual effort, research has shown that they do not reliably lead to better collaboration in teams (Riedl & Woolley, 2017). However, paying for individual performance (or payments based on individual process metrics) in the context of collaborative work risks undermining group cooperation and the quality of group output (Wageman & Baker, 1997). There is also evidence suggesting that incentives based on effort alone can lead participants to game the system in order to maximize their “effort” without enhancing work quality (Ordóñez, et al., 2009).

The effects of some of these types of incentives also depend on other dimensions. For instance, while the *Feedback Display* nudge we tested here had unintended effects, a prior study testing an intervention very similar to the one tested here found that it had a positive effect in distributed student teams comprised of team members who were familiar with each other collaborating on a semester-long project (Glikson et al., 2019). We interpret this as suggesting that the nudge can spur action in teams comprised of members who know each other or have a preexisting relationship. Even some minimal “reputational stake” changes the behavior of low-effort contributors (Hong et al., 2018; Lu et al., 2017; Resnick et al., 2000), as there is considerable evidence highlighting conformity and reciprocity pressures as feasible mechanisms for increasing effort when others are contributing to a common cause (Bardsley & Sausgruber, 2005). Such a reputational effect on effort can be tested by comparing the results of the current experiment with the performance of teams of close friends or teams of strangers who are likely to engage in repeated future interactions.

Taken together, we postulate that the effects of the nudge treatments observed in this experiment should generalize to other temporary and anonymous team populations and could be more effective in samples that are different on any one of these

dimensions, as it could increase the conformity or reciprocity pressure needed to increase participant effort. In populations that contradict most or all of these dimensions (i.e., preexisting teams with familiar members), the efficacy of our facilitator bot nudges will depend on how well members know each other and how well they have worked together on similar tasks in the past, as these nudges are most helpful to teams who lack information about each other and/or about how to work together on the task. With respect to incentives, we conclude that individual-level motivators are unlikely to work out across all populations and nudge types, while motivating through reputational stakes could be effective when engaging the more dynamic nudges of *Feedback Display* and *ToDo List*.

Conclusion

In conclusion, though our experiment yielded a mix of predicted and unexpected results, our findings suggest a role for digital nudges, particularly those utilizing more sophisticated artificial social intelligence, to enhance collective intelligence. We noted some of the opportunities to improve our failed nudge designs by adapting based on collaborators' behavior. Taking that a step further, we believe it would be possible to build on the modest success of our *Skill Facilitator* by incorporating a more intelligent agent who could coordinate inputs prompted by other nudges to adaptively "coach" collaborators on how to use information about each other's skills to devise an effective task strategy and connect to members who feel disengaged. However, such attempts should proceed with the same cautions that surfaced in our structured speculation, attending to collective behavior, the overall "dose" of the treatment, and the population characteristics. That said, if calibrated correctly, such interventions offer an exciting possibility to help collaborators quickly develop the various forms of collective cognition that research shows are important to the quality of collaboration. Such interventions could thus lead to tremendous opportunities to cultivate collective intelligence in temporary online groups and enable us to effectively address an even larger set of our most complex problems.

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